Performance Comparison of Different Machine Learning Techniques for Early Prediction of Breast Cancer using Wisconsin Breast Cancer Dataset

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***Abstract*— A significant health issue, cancer is becoming more prevalent globally and is a leading cause of mortality. Recent studies have shown that breast cancer is one of the most prevalent cancer type, particularly among women. Early detection can increase the chances of survival for those with breast cancer and lower treatment cost. However, there are drawbacks to the early diagnosis methods utilized in today's healthcare systems. These include the need for substantial human resources, long-term effects, and difficult access to these services for everybody. For early breast cancer diagnosis, technologies that are simple to use, yield reliable findings compared to scientific methodologies, and are available to everyone are required. Artificial Intelligence techniques enable the early diagnosis of breast cancer. This study aims to classify benign and malignant breast cancer image features. Artificial Neural Network, Support Vector Machine and Random Forest algorithms were used to classify features obtained from images. Experiments were performed on the Wisconsin Breast Cancer dataset. Experimental evaluation shows that 99% of the most successful results were achieved with the Artificial Neural Network algorithm. According to experimental findings, the classification technique can identify breast cancer in its early stages. The findings of the study are expected to shed on light new researches for investigation into breast cancer early detection.**

Keywords*— *Breast cancer, early prediction, machine learning, wisconsin breast cancer dataset

I. Introduction

The most lethal illness, cancer, is also the one that is expanding over the globe. Cancer, defined as the uncontrolled development of various diseases throughout the body, is a non-lethal "genetic condition." In the human body, cancer can cause severe degeneration and death that might be irreversible. Studies show that one out of every six deaths globally are due to cancer. In particular, 70% of deaths happening in less and less-developed countries are caused by cancer. Furthermore, according to the data from the World Health Organization (2020), it is seen that the types of breast, lung and large intestine cancers have increased in recent years. These types of cancer account for half of all cancer cases worldwide and are common among women. In addition, breast cancer corresponds to 30% of new cancer cases diagnosed in women [1]. Therefore, determining breast cancer's harmful effects on people's survival and early diagnosis of this disease is essential for prolonging the lifespan. Early detection and diagnosis methods are often used to help breast cancer patients recover or extend their survival [2]. Techniques including mammography, ultrasonography, magnetic resonance imaging, and the gentlest identify breast cancer, but when the disease is advanced, the outcome of this detection places a heavy financial load on the patient.

In this study, we focus on applying classification algorithms for early breast cancer prediction. The approach most frequently used in data mining applications for health analytics is classification. It classifies data or records using classification rules. This paper aims to conduct analysis on a known breast cancer dataset, "Wisconsin Breast Cancer Dataset," using a classification data mining techniques for early disease prediction. In order to select the optimum classification technique, we first classify using the Artificial Neural Network, Support Vector Machine, and Random methods.

The principal contributions of this paper are:

- Choosing the higher accuracy performance of ML algorithms for early predicting of breast cancer.
- Assessing performance of various machine learning algorithms on breast cancer dataset.

The paper's organization is arranged as follows: Related studies in the literature are analyzed in Section II. The details of the Methodology are described in Section III. The experimental analysis of the comparative algorithms is shown in Section IV. Finally, section V will benchmark, discuss research results, and give recommendations for future research directions.

II. RELATED WORK

There are several models and strategies that employ various imaging modalities to predict breast cancer early in the literature. Mammography is the most basic method preferred for imaging the breast. It is the approach that, out of all imaging and diagnostic methods, may find breast cancer at the earliest stage. Therefore, mammography aims to detect breast cancer at a very early phase of the disease. To take an image in this technique, the breast image must be slightly compressed between the image reception device and the compression plate. In standard mammography, two divergent pose images are taken from the front and side of each breast.

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Other films may be required to visualize and diagnose this area better if there is a doubtful result. Ultrasonography, another method used for detecting breast cancer, is a method that provides images through sound waves. Ultrasonography, unlike mammography, is a cross-sectional imaging method. An ultrasound examination of the breast is recommended in women with dense breast tissue. In this way, small cancer foci covered by breast tissue in mammography can be detected more easily by ultrasound examination. The Magnetic Resonance Imaging (MR) method is the most sensitive to diagnosing breast cancer. It determines the extent of detected cancer when a suspicious finding is detected in mammography or ultrasonography. In the gentlest method, the relational analysis of genes related to breast cancer is performed. These genes aim to ensure the normal development of breast cells and prevent the proliferation of cancer cells [3-5].

Use of the procedures, access to the equipment, and the facilities where the devices are offered, particularly the field knowledge, are all prerequisites. Furthermore, it might be difficult to treat all patients with breast cancer promptly at healthcare facilities and to continue imaging research on at-risk patient populations or age groups [6]. These factors raise the demand for tools to help dermatologists and patients with issues like hospital access to early breast cancer detection.

The constraints made innovative methods possible, such as the lengthy imaging processes required for early identification of breast cancer. The increasing use of AI techniques in the health industry is a result of the digital revolution of this industry. With the help of AI techniques, the speed and transaction costs of many works offered in the health industry have improved [7-9]. The benefits of AI have also been incorporated into breast cancer imaging methods. The potential of image processing techniques in particular has created chances for the early detection of breast cancer utilizing AI and ML techniques. The literature is seeing an increase in studies utilizing AI and ML methods for breast cancer early detection. It is seen that intelligent systems developed in research help facilitate the work of doctors and patients. Comprehensive images and features extracted from images are imperative for the early detection of breast cancer with AI techniques. The research's image characteristics and content quality are both closely connected to the classification performance of the selected algorithms [10]. In this approach, it is essential to choose data sets that can produce precise findings using thorough and scientific techniques in the classification procedures for the early identification of breast cancer [11-16].

Table I contains a summary of information about the research conducted on the data set.

TABLE I. SUMMARY OF PREVIOUS STUDIES

Reference	Aim	Methods	Best Accuracy Performance
[17]	To evaluate the performance of different ML techniques used for the early prediction of breast cancer	Decision Tree 4.5, Simple Logistic, KNN, SVM, NB	SVM, KNN, Simple Logistic: 98.1%
[18]	To conduct performance and comparison analysis on ML techniques for the Breast Cancer Wisconsin Dataset	KNN, SVM, Decision Tree 4.5, Random Forest, Logistic Regression	SVM: 97.2%
[19]	To compare ML algorithms by measuring their accuracy, sensitivity, and specificity values for Wisconsin Diagnostic Dataset	Multilayer Perceptron, GRU-SVM, Softmax Regression, Linear Regression, KNN, SVM	Multilayer Perceptron: 99.03%
[20]	To investigate how different feature selection methods affect the Wisconsin Diagnostic Dataset's ML algorithms' accuracy	Random Forest, Logistic Regression, Naïve Bayes	Random Forest: 97.14%
[21]	To investigate Azure machine learning (AzureML) for breast cancer early detection	Decision Forest Decision Jungle	Decision Jungle 97%
[22]	To carry out ML techniques for early prediction of breast cancer	Gradient Boosted Decision Tree, Generalized Linear Model, Logistic Regression	Generalized Linear Model 99.4%
[23]	To predict breast cancer using several ML techniques	Random Forest, Naïve Bayes, SVM, KNN	SVM: 97.9%
[24]	To predict breast cancer, developing a Deep Neural Network	Deep Neural Network	Deep Neural Network: 99.42%
[25]	To conduct a comparative study of the ML algorithms for breast cancer	Naïve Bayes, Random Forest, KNN	KNN: 95.90%
[26]	To demonstrate a breast cancer prediction using several machine learning algorithms and their classification outcomes	Random Forest, Naïve Bayes, Logistic Regression, SVM, KNN	SVM: 96.25%
[27]	To predict breast cancer using divergent ML techniques	SVM, Random Forest, KNN	SVM: 98.25%

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[28]	To evaluate the accuracy of several ML methods for breast cancer prediction	KNN, SVM, Naïve Bayes	KNN: 96.85%
[29]	To predict breast cancer in the context of malignant or benign tumors	KNN, SVM, AdaBoost	KNN: 99.12%
[30]	To compare the effectiveness of ML techniques for breast cancer early detection	Logistic Regression, Decision Tree	Decision Tree 95%
[31]	To compare classification performances of ML techniques for predicting breast cancer	Naive Bayes (NB), SVM, Logistic Regression (LR), KNN	Logistic Regression: 99%
[32]	To compare performances of various ML techniques on the Wisconsin Dataset	SVM, Random Forest, Decision Tree 4.5	SVM: 97.66%
[33]	To compare and assess the effectiveness of ML methods used for breast cancer early prediction	Bagging, Random Forest, AdaBoost, SVM, KNN	AdaBoost: 98.77%
[34]	To propose Deep Learning Models for the early prediction of breast cancer	Naïve Bayes, Decision Tree, SVM, Vote, Random Forest, AdaBoost	Deep Learning (ELU): 96.99 %
[35]	To conduct a performance analysis on Wisconsin Dataset using ML for predicting breast cancer	Decision Tree, Naïve Bayes, SVM, Random Forest, Logistic Regression	Random Forest, SVM: 95%
[36]	To propose a framework in the context of Big Data for predicting Breast Cancer	Decision Tree, Random Forest	Random Forest 93.64%
[37]	To predict breast cancer using ML techniques	Tree, Naïve Bayes, Linear Discriminant Analysis	SVM: 95%

III. METHODOLOGY

This section describes the methodological processes of the research.

This study uses the Wisconsin Breast Dataset to propose classification methods for the early diagnosis of breast cancer. The data was classified using the Artificial Neural Network, Support Vector Machine, and Random Forest techniques.

The flow chart in Figure 1 describes the steps followed for the classification process.

Fig. 1. Flowchart of Study

A. Data Description

There are several public datasets available that contain images of breast cancer. In this study, we used Wisconsin Breast Cancer Dataset the Wisconsin from UCI ML Repository. This dataset consists of 699 instances with 11 attributes. Table II contains the features of the data set.

TABLE II. FEATURES RELATED TO DATASET

No	Feature Title	Description
1	Radius (Average distances between points on the perimeter and the center)	Average distances between points on the perimeter and the center
2	Texture (Gray-Scale Values' Standard Deviation)	The grayscale values' standard deviation
3	Perimeter	Typical tumor core size
4	Area	
5	Smoothness (Radius length variations on a local level)	Average of local variation in radius lengths
6	Concavity (Concave portions of the contour's severity)	The standard error for perimeter^2 / area - 1.0
7	Concave Points (The proportion of the contour's concave areas)	Average for the proportion of the contour that is concave
8	Symmetry	
9	Fractal Dimension ("Coastline Approximation" — 1)	Average for "coastline approximation" – 1

B. Data Preprocessing

Datasets contain many types of properties and values. For example, data can include missing values, duplicated data, or noisy values. In this situation, data preprocessing must be performed. Therefore, reducing lost values and noisy data in data preprocessing is necessary.

Processes such as data preprocessing, data standardization, and categorical data conversion into numerical data was carried out. Data standardization creates a standard data format from different data sets. After the data are gathered from various sources and before they are included in the targeted research subject, it deals with the transformation of datasets.

We used the sklearn.preprocessing.StandardScaler structure of the sklearn library to standardize the data. Using the fit_transform() method inside the StandardScaler class, we compressed data with very large values and points with very small values between a certain number.

C. Machine Learning Algorithms

The model was created using machine learning algorithms such the Artificial Neural Network, Support Vector Machine, and Random Forest.

ANN are mathematical models composed of many processing units (neurons) linked together in a weighted manner. For example, ANN's calculation order begins with

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receiving signals from other neurons, combining them, transforming them and ends with producing a numerical result. ANNs work distinctively from standard processors. While each task is carried out by a single central processing unit in conventional processors, each simple processing unit in an ANN handles a different aspect of a complicated problem.

Support vector machines (SVMs) are vector space-based machine learning techniques that determine which class is farthest away from the other in training data. Data from binary classification are primarily discriminated using SVM.

The Random Forest algorithm is one of the supervised classification algorithms. It is used in both classification problems and regression. The technique generates many decision trees in order to improve classification performance during computation. A random forest algorithm selects the decision tree with the highest score out of numerous ones that operate independently of one another [38].

D. Evaluation Metrics

We used precision, recall, F-measure, and accuracy values presented in Table III. Precision is the proportion of estimated positive values. Recall value is a metric that shows how many transactions need to predict as positive. The harmonic mean of recall and accuracy is known as the F-measure. The most used metric for accuracy is the proportion of properly identified samples to the total number of samples.

E. Experimental Evaluation

This section discusses the "Wisconsin Breast Cancer" dataset, experimental results, and evaluation metrics. In this research, we use Python 3.10 to implement our proposed method.

We tested our classification technique using the k - fold cross-validation method. In addition, we have utilized the 10-fold technique: the data set is separated into ten chunks. Nine of the dataset's folds are utilized for training, while the last set is used for analysis and testing. Out of the 569 examples, we used 398 for the training set and 171 for testing.

TABLE III. CALCULATION METRICS USED FOR EXPERIMENTAL EVALUATION

Technique	Formula
Accuracy	$\left(\frac{(TP+TN)}{TP+FP+TN+FN}\right)$
Recall	TP/(TP + FN)
Precision	TP/(TP+TN)

We assigned different values hyperparameters for selected algorithms.

The kernel value for the SVM algorithm is determined as "rbf," and its degree is 3.

Except for the last layer for the ANN algorithm, the activation functions of the other Fully Connected Dense layers were determined as "ReLu" and the activation function as "sigmoid". The model's loss function was set to "Sparse Categorical Cross Entropy," optimizer "RMSprop," batch-size 1, epoch 10, and these other parameters.

For the Random Forest algorithm, the application was carried out using parameters such as n_estimators (The number of trees in the forest) value of 100 and criterion value "gini".

IV. RESULTS

This section illustrates the results of the experimental evaluation. Table IV shows the results of the performance criteria.

TABLE IV. DETAILS ABOUT PERFORMANCE CRITERIA AND RESULTS

Algorithm	Acc	Auc	Precison	Recall	F-Sc
SVM	0.97	0.97	0.97	0.98	0.98
ANN	0.99	0.99	0.99	0.98	0.99
Random Forest	0.97	0.96	0.98	0.99	0.97

V. CONCLUSION

Breast cancer is one of the most common cancers in women and early detection is crucial to assisting medical professionals in their job and extending the lives of patients. This study uses the Wisconsin Breast Cancer Dataset to compare several machine learning methods for identifying breast cancer. In general, results show that each algorithm had more than 90% accuracy in determining benign or malignant tumors. Furthermore, the comparative results indicate that the Artificial Neural Network achieves the highest prediction performance among the three algorithms. The use of it for breast cancer prediction so seems to be quite appropriate.

Several recent studies [38] performed classification using Wisconsin Breast Cancer Dataset with 31 features and 569 samples by KNN, RF, SVM, ANN, Linear Regression, naïve Bayes, and Gradient Booster algorithms, obtaining 98.83% accuracy with the Random Forest algorithm. [39] got 97% accuracy with SVM by classifying with Decision Tree, SVM, KNN and Random Forest on the same data set. [40]made classification using Decision Tree and ANN and achieved 98% accuracy with ANN. [41] obtained 96.5% accuracy using Random Forest and SVM algorithms. Finally, [37] classified with SVM, Decision Tree, Naive Bayes, Logistic Regression and KNN and achieved 95% accuracy with SVM. [42] performed Multilayer Perceptron, Naïve Bayes, Random Forest, and Simple Logistic Regression classification on the Coimbra dataset and obtained 85% accuracy with Multilayer Perceptron.

In the context of SVM and Random Forest algorithms, the results produced within the study's parameters are observed to be superior to the performances of previous studies on the data set when compared to research results in the literature. Furthermore, the research results show that the performance results are high compared with related studies on different data sets. The comparisons show that the research results and the research results in the literature are consistent. Therefore, a possible direction for future studies might be to test the results with different features and other machine learning

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techniques and develop mobile applications that instantly display the classification results.

REFERENCES

[1] M. B. Siegel *et al.*, "Integrated RNA and DNA sequencing reveals early drivers of metastatic breast cancer," *Journal of Clinical Investigation*, vol. 128, no. 4, pp. 1371–1383, Apr. 2018, doi: 10.1172/JCI96153.

[2] J. J. Ott, A. Ullrich, and A. B. Miller, "The importance of early symptom recognition in the context of early detection and cancer survival," *European Journal of Cancer*, vol. 45, no. 16, pp. 2743–2748, Nov. 2009, doi: 10.1016/j.ejca.2009.08.009.

[3] J. G. Elmore, "Screening for Breast Cancer," *JAMA*, vol. 293, no. 10, p. 1245, Mar. 2005, doi: 10.1001/jama.293.10.1245.

[4] A. K. Das, S. Kr. Biswas, A. Bhattacharya, and E. Alam, "Introduction to Breast Cancer and Awareness," in *2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Mar. 2021, pp. 227–232. doi: 10.1109/ICACCS51430.2021.9441686.

[5] Y.-S. Sun *et al.*, "Risk Factors and Preventions of Breast Cancer," *International Journal of Biological Sciences*, vol. 13, no. 11, pp. 1387–1397, 2017, doi: 10.7150/ijbs.21635.

[6] R. Krithiga and P. Geetha, "Breast Cancer Detection, Segmentation and Classification on Histopathology Images Analysis: A Systematic Review," *Archives of Computational Methods in Engineering*, vol. 28, no. 4, pp. 2607–2619, Jun. 2021, doi: 10.1007/s11831-020-09470-w.

[7] R. Buettner, M. Bilo, N. Bay, and T. Zubac, "A Systematic Literature Review of Medical Image Analysis Using Deep Learning," in *2020 IEEE Symposium on Industrial Electronics & Applications (ISIEA)*, Jul. 2020, pp. 1–4. doi: 10.1109/ISIEA49364.2020.9188131.

[8] A. Rehman, M. Ahmed Butt, and M. Zaman, "A Survey of Medical Image Analysis Using Deep Learning Approaches," in *2021 5th International Conference on Computing Methodologies and Communication (ICCMC)*, Apr. 2021, pp. 1334–1342. doi: 10.1109/ICCMC51019.2021.9418385.

[9] V. Malpe and P. Tugaonkar, "Machine LearningTrends in Medical Sciences," in *2018 2nd International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), 2018 2nd International Conference on*, Aug. 2018, pp. 495–499. doi: 10.1109/I-SMAC.2018.8653756.

[10] P. H. Abreu, M. S. Santos, M. H. Abreu, B. Andrade, and D. C. Silva, "Predicting Breast Cancer Recurrence Using Machine Learning Techniques," *ACM Computing Surveys*, vol. 49, no. 3, pp. 1–40, Dec. 2016, doi: 10.1145/2988544.

[11] S. C. S R and H. Rajaguru, "A Systematic Review on Screening, Examining and Classification of Breast Cancer," in *2021 Smart Technologies, Communication and Robotics (STCR)*, Oct. 2021, pp. 1–4. doi: 10.1109/STCR51658.2021.9588828.

[12] M. Vasudev, A. Doegar, and V. Gupta, "A Review on Automatic Classification of Breast Cancer Using Supervised Learning Strategies," in *2019 4th International Conference on Information Systems and Computer Networks (ISCON)*, Nov. 2019, pp. 717–721. doi: 10.1109/ISCON47742.2019.9036261.

[13] A. Kajala and V. K. Jain, "Diagnosis of Breast Cancer using Machine Learning Algorithms-A Review," in *2020 International Conference on Emerging Trends in Communication, Control and Computing (ICONC3)*, Feb. 2020, pp. 1–5. doi: 10.1109/ICONC345789.2020.9117320.

[14] B. Jaafar, H. Mahersia, and Z. Lachiri, "A survey on deep learning techniques used for breast cancer detection," in *2020 5th International Conference on Advanced Technologies for Signal and Image Processing (ATSIP)*, Sep. 2020, pp. 1–6. doi: 10.1109/ATSIP49331.2020.9231684.

[15] S. Nadkarni and K. Noronha, "A Review on Multiparametric Magnetic Resonance Imaging for the Detection of Breast Cancer," in *2020 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, Jul. 2020, pp. 1–5. doi: 10.1109/CONECCT50063.2020.9198362.

[16] A. Saber, M. Sakr, O. M. Abo-Seida, and A. Keshk, "Automated Breast Cancer Detection and Classification Techniques – A survey," in *2021 International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC)*, May 2021, pp. 200–207. doi: 10.1109/MIUCC52538.2021.9447655.

[17] M. A. Naji, S. el Filali, M. Bouhlal, E. H. Benlahmar, R. A. Abdelouhahid, and O. Debauche, "Breast Cancer Prediction and Diagnosis through a New Approach based on Majority Voting Ensemble Classifier," *Procedia Computer Science*, vol. 191, pp. 481–486, 2021, doi: 10.1016/j.procs.2021.07.061.

[18] M. A. Naji, S. el Filali, K. Aarika, E. H. Benlahmar, R. A. Abdelouhahid, and O. Debauche, "Machine Learning Algorithms For Breast Cancer Prediction And Diagnosis," *Procedia Computer Science*, vol. 191, pp. 487–492, 2021, doi: 10.1016/j.procs.2021.07.062.

[19] A. F. M. Agarap, "On breast cancer detection," in *Proceedings of the 2nd International Conference on Machine Learning and Soft Computing - ICMLSC '18*, 2018, pp. 5–9. doi: 10.1145/3184066.3184080.

[20] R. Dhanya, I. R. Paul, S. Sindhu Akula, M. Sivakumar, and J. J. Nair, "A Comparative Study for Breast Cancer Prediction using Machine Learning and Feature Selection," in *2019 International Conference on Intelligent Computing and Control Systems (ICCS)*, May 2019, pp. 1049–1055. doi: 10.1109/ICCS45141.2019.9065563.

[21] K. Alshouiliy, A. Shivanna, S. Ray, A. AlGhamdi, and D. P. Agrawal, "Analysis and Prediction of Breast Cancer using AzureML Platform," in *2019 IEEE 10th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Oct. 2019, pp. 0212–0218. doi: 10.1109/IEMCON.2019.8936294.

[22] R. Fitrah Umami and R. Sarno, "Analysis of Classification Algorithm for Wisconsin Diagnosis Breast Cancer Data Study," in *2020 International Seminar on Application for Technology of Information and Communication (iSemantic)*, Sep. 2020, pp. 464–469. doi: 10.1109/iSemantic50169.2020.9234295.

[23] Y. Khourdifi and M. Bahaj, "Applying Best Machine Learning Algorithms for Breast Cancer Prediction and Classification," in *2018 International Conference on Electronics, Control, Optimization and Computer Science (ICECOCS)*, Dec. 2018, pp. 1–5. doi: 10.1109/ICECOCS.2018.8610632.

[24] Md. O. F. Goni, F. Md. S. Hasnain, Md. A. I. Siddique, O. Jyoti, and Md. H. Rahaman, "Breast Cancer Detection using Deep Neural Network," in *2020 23rd International Conference on Computer and Information Technology (ICCIT)*, Dec. 2020, pp. 1–5. doi: 10.1109/ICCIT51783.2020.9392705.

[25] S. Sharma, A. Aggarwal, and T. Choudhury, "Breast Cancer Detection Using Machine Learning Algorithms," in *2018 International Conference on Computational Techniques, Electronics and Mechanical Systems (CTEMS)*, Dec. 2018, pp. 114–118. doi: 10.1109/CTEMS.2018.8769187.

[26] V. A. Telsang and K. Hegde, "Breast Cancer Prediction Analysis using Machine Learning Algorithms," in *2020 International Conference on Communication, Computing and Industry 4.0 (C2I4)*, Dec. 2020, pp. 1–5. doi: 10.1109/C2I451079.2020.9368911.

[27] U. Pratap and S. Chhabra, "Breast Cancer Prediction using Different Machine Learning Algorithms," in *2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, Dec. 2021, pp. 451–454. doi: 10.1109/ICAC3N53548.2021.9725688.

[28] B. Akbugday, "Classification of Breast Cancer Data Using Machine Learning Algorithms," in *2019 Medical Technologies Congress (TIPTEKNO)*, Oct. 2019, pp. 1–4. doi: 10.1109/TIPTEKNO.2019.8895222.

[29] S. Laghmati, B. Cherradi, A. Tmiri, O. Daanouni, and S. Hamida, "Classification of Patients with Breast Cancer using Neighbourhood Component Analysis and Supervised Machine Learning Techniques," in *2020 3rd International Conference on Advanced Communication Technologies and Networking (CommNet)*, Sep. 2020, pp. 1–6. doi: 10.1109/CommNet49926.2020.9199633.

[30] P. P. Sengar, M. J. Gaikwad, and A. S. Nagdive, "Comparative Study of Machine Learning Algorithms for Breast Cancer Prediction," in *2020 Third International Conference on Smart Systems and Inventive Technology (ICSSIT)*, Aug. 2020, pp. 796–801. doi: 10.1109/ICSSIT48917.2020.9214267.

[31] B. R. Roy, M. Pal, S. Das, and A. Huq, "Comparative Study of Machine Learning Approaches on Diagnosing Breast Cancer for Two Different Dataset," in *2020 2nd International Conference on Advanced Information and Communication Technology (ICAICT)*, Nov. 2020, pp. 29–34. doi: 10.1109/ICAICT51780.2020.9333507.

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[32] J. Ivancakova, F. Babic, and P. Butka, "Comparison of different machine learning methods on Wisconsin dataset," in *2018 IEEE 16th World Symposium on Applied Machine Intelligence and Informatics (SAMI)*, Feb. 2018, pp. 000173–000178. doi: 10.1109/SAMI.2018.8324834.

[33] N. A. Mashudi, S. A. Rossli, N. Ahmad, and N. M. Noor, "Comparison on Some Machine Learning Techniques in Breast Cancer Classification," in *2020 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES)*, Mar. 2021, pp. 499–504. doi: 10.1109/IECBES48179.2021.9398837.

[34] P. Mekha and N. Teeyasuksaet, "Deep Learning Algorithms for Predicting Breast Cancer Based on Tumor Cells," in *2019 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT-NCON)*, Jan. 2019, pp. 343–346. doi: 10.1109/ECTI-NCON.2019.8692297.

[35] G. Gupta, M. Sharma, S. Choudhary, and K. Pandey, "Performance Analysis of Machine Learning Classification Algorithms for Breast Cancer Diagnosis," in *2021 9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, Sep. 2021, pp. 1–6. doi: 10.1109/ICRITO51393.2021.9596230.

[36] S. Ray, A. AlGhamdi, A. AlGhamdi, K. Alshouiliy, and D. P. Agrawal, "Selecting Features for Breast Cancer Analysis and Prediction," in *2020 International Conference on Advances in Computing and Communication Engineering (ICACCE)*, Jun. 2020, pp. 1–6. doi: 10.1109/ICACCE49060.2020.9154919.

[37] A. Ivaturi, A. Singh, B. Gunanvitha, and K. S. Chethan, "Soft Classification Techniques for Breast Cancer Detection and Classification," in *2020 International Conference on Intelligent Engineering and Management (ICIEM)*, Jun. 2020, pp. 437–442. doi: 10.1109/ICIEM48762.2020.9160219.

[38] K. Mridha, "Early Prediction of Breast Cancer by using Artificial Neural Network and Machine Learning Techniques," in *2021 10th IEEE International Conference on Communication Systems and Network Technologies (CSNT)*, Jun. 2021, pp. 582–587. doi: 10.1109/CSNT51715.2021.9509658.

[39] C. Kaul and N. Sharma, "High Accuracy Predictive Model on Breast Cancer Using Ensemble Approach of Supervised Machine Learning Algorithms," in *2021 International Conference on Computational Performance Evaluation (ComPE)*, Dec. 2021, pp. 071–076. doi: 10.1109/ComPE53109.2021.9752254.

[40] R. Hazra, M. Banerjee, and L. Badia, "Machine Learning for Breast Cancer Classification With ANN and Decision Tree," in *2020 11th IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Nov. 2020, pp. 0522–0527. doi: 10.1109/IEMCON51383.2020.9284936.

[41] S. Ara, A. Das, and A. Dey, "Malignant and Benign Breast Cancer Classification using Machine Learning Algorithms," in *2021 International Conference on Artificial Intelligence (ICAI)*, Apr. 2021, pp. 97–101. doi: 10.1109/ICAI52203.2021.9445249.

[42] T. Khatun *et al.*, "Performance Analysis of Breast Cancer: A Machine Learning Approach," in *2021 Third International Conference on Inventive Research in Computing Applications (ICIRCA)*, Sep. 2021, pp. 1426–1434. doi: 10.1109/ICIRCA51532.2021.9544879.

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